Course Notes

for (CSED490V) Foundations of Continuous Computation

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Notation and Conventions

General notation and writing follow *(MATH000) Mathematical Language*.

Notation	Meaning
$\mathbb{R}_C$	Computable real numbers
P_{root}	Integer polynomials admitting an integer root
$\mathcal{A}$	Oracle algorithm
$\gamma_{E,M}$	Floating-point representation map
$\mathbb{F}_{E,M}$	Finite floating-point values
$\overline{\mathbb{F}}_{E,M}$	Extended floating-point values
$e_{\min}$	Minimum normal exponent
$e_{\max}$	Maximum normal exponent
$F_{\max}$	Largest finite floating-point value
$K_{E,M}$	Overflow threshold
$\text{rnd}_{E,M}$	Round to nearest, ties to even
$\oplus$	Rounded addition
$\ominus$	Rounded subtraction
$\otimes$, $\odot$	Rounded multiplication
$\oslash$, $\oslash$	Rounded division
fma	Fused multiply-add
err_{abs}	Absolute error
err_{rel}	Relative error
ulp	Unit in the last place
err_{ulp}	ULP error
e_a	Exponent of a nonzero floating-point value a
fast2sum	Error-free summation algorithm
σ	Input valuation
Σ	Set of admissible input valuations
$[\![e]\!]$	Floating-point semantics of e
$[\![e]\!]_{\text{r}}$	Real-valued semantics of e
$\text{range}(e)$	Range enclosure over Σ
$\text{err}(e)$	Absolute-error bound over Σ
ε	Machine epsilon (unit roundoff)

1 Computable Reals

Definition 1.1. Borel Computability
A real number r is *Borel-computable* if there exists a computable sequence $(q_n)_{n\in\mathbb{N}_0} \subseteq \mathbb{Q}$ such that

$$|r - q_n| < 2^{-n} \qquad (n \in \mathbb{N}_0).$$

Definition 1.2. Turing Computability
A nonnegative real number r is *Turing-computable* if it has a binary expansion

$$r = a + \sum_{i=0}^{\infty} b_i 2^{-(i+1)}, \qquad a \in \mathbb{N}_0, \quad b_i \in \{0, 1\},$$

for which $i \mapsto b_i$ is computable. A negative real number is Turing-computable when its absolute value is Turing-computable. For dyadic rationals we may use the terminating binary expansion.

Theorem 1.3. Equivalence of Computability Definitions
For $r \in \mathbb{R}$, the following are equivalent:

(i) r is Borel-computable.

(ii) r is Turing-computable.

Proof. We may consider $r \geq 0$: changing the sign of a computable rational approximation is effective, and the negative case of binary computability is defined using the absolute value.

Suppose first that r is Turing-computable, with integer part a and computable bits b_i. Set

$$q_n = a + \sum_{i=0}^{n} b_i 2^{-(i+1)} \in \mathbb{Q}.$$

The sum is computable and

$$0 \leq r - q_n \leq \sum_{i=n+1}^{\infty} 2^{-(i+1)} = 2^{-(n+1)} < 2^{-n}.$$

Thus r is Borel-computable.

Conversely, suppose $(q_m)_{m \in \mathbb{N}_0}$ computes r with $|q_m - r| < 2^{-m}$. Fix the integer part $a = \lfloor r \rfloor$ as part of the existential description of a binary expansion. We distinguish two cases.

If r is dyadic, it admits an eventually zero binary expansion. Its finite sequence of nonzero digits can be built into an algorithm, and hence r is Turing-computable.

Now assume r is not dyadic. Fix $n \in \mathbb{N}_0$. The non-dyadic number r lies strictly between two consecutive grid points of mesh $2^{-(n+1)}$. Search over $m \in \mathbb{N}_0$ until the rational interval

$$I_m = (q_m - 2^{-m},\, q_m + 2^{-m})$$

is contained strictly in a single grid cell

$$\left(\frac{k}{2^{n+1}}, \frac{k+1}{2^{n+1}} \right), \qquad k \in \mathbb{N}_0.$$

Containment is decidable by rational arithmetic: for a given m there is at most one candidate k, which can be found by integer division. The search terminates, since r has positive distance from both endpoints of its grid cell and $q_m \to r$ with an effective error bound. Moreover $r \in I_m$, so the returned k satisfies $k = \lfloor 2^{n+1} r \rfloor$. The nth binary digit of r is therefore

$$b_n = k \bmod 2.$$

This procedure computes each bit, completing the proof. □

Remark. The two definitions characterize the same set of real numbers, but their representations do not admit symmetric *uniform* conversions. Given a computable binary expansion one can effectively obtain rational approximations. In general there is no algorithm that takes an arbitrary rational-approximation procedure and uniformly produces its binary digits: the latter operation must resolve boundary cases at dyadic rationals. This does not contradict the existence of a computable binary expansion for each Borel-computable real.

Definition 1.4. Computable Real Numbers
The set of *computable real numbers* is

$$\mathbb{R}_C := \{r \in \mathbb{R} \mid r \text{ is computable}\}.$$

By the preceding theorem, either notion of computability can be used.

Proposition 1.5. Closure under Arithmetic
The set $\mathbb{R}_C$ is a subfield of $\mathbb{R}$. In particular,

$$x, y \in \mathbb{R}_C \implies x+y,\ x-y,\ xy \in \mathbb{R}_C, \qquad y \neq 0 \implies \frac{x}{y} \in \mathbb{R}_C.$$

Proposition 1.6. Real Closedness of Computable Reals
Every real root of a nonzero polynomial in $\mathbb{R}_C[X]$ is computable. Consequently, $\mathbb{R}_C$ is a real-closed field.

Proposition 1.7. Computability of π and e
The real numbers π and e are computable.

Proof. The alternating arctangent series gives

$$\frac{\pi}{4} = \arctan 1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}.$$

For $m \in \mathbb{N}_0$ put

$$p_m = 4 \sum_{k=0}^{m} \frac{(-1)^k}{2k+1} \in \mathbb{Q}.$$

The alternating-series estimate yields

$$|\pi - p_m| \leq \frac{4}{2m+3}.$$

Choosing $m = 2^{n+1}$ gives $|\pi - p_m| < 2^{-n}$ for every $n \in \mathbb{N}_0$. The finite sum p_m is computable from n.

The computability of $e = \exp(1)$ follows from the computability of $\exp\colon \mathbb{R} \to \mathbb{R}$ proved in Theorem 2.8. □

Proposition 1.8. Countability of the Computable Reals
The set $\mathbb{R}_C$ is countably infinite. Consequently, there exist noncomputable real numbers.

Proof. There are countably many finite descriptions of algorithms, and every computable real has at least one such description. Hence $\mathbb{R}_C$ is at most countable. On the other hand, every rational number is computable via its constant approximation sequence, so $\mathbb{Q} \subseteq \mathbb{R}_C$ and $\mathbb{R}_C$ is infinite. Since $\mathbb{R}$ is uncountable, $\mathbb{R} \setminus \mathbb{R}_C \neq \varnothing$. □

Fact 1.9. Hilbert's Tenth Problem
There is no algorithm that decides, for every multivariate polynomial with integer coefficients, whether it has an integer zero. In particular, the set of integer polynomials admitting an integer solution is not decidable.

Proposition 1.10. A Noncomputable Real
Let P be the set of all integer-coefficient polynomials in finitely many variables, and let $\phi\colon P \to \mathbb{N}$ be a computable injection. Write $P_{\text{root}} \subseteq P$ for the polynomials having an integer zero. Then

$$r \coloneqq \sum_{p \in P} 2^{-\phi(p)}\, \mathbb{1}_{P_{\text{root}}}(p)$$

is a noncomputable real in $[0,1]$.

Proof. Injectivity gives the binary coefficient at $2^{-\phi(p)}$:

$$b_{\phi(p)} = \mathbb{1}_{P_{\text{root}}}(p).$$

Both P_{root} and $P \setminus P_{\text{root}}$ are infinite, so the expansion has infinitely many zeros and ones; in particular, r is not dyadic. If $r \in \mathbb{R}_C$, its unique binary expansion is computable by the equivalence theorem. Hence

$$p \longmapsto b_{\phi(p)} = \mathbb{1}_{P_{\text{root}}}(p)$$

would decide Hilbert's Tenth Problem, a contradiction. □

Remark. The construction is not specific to P_{root}. If $A \subseteq \mathbb{N}_0$ is noncomputable, then

$$r_A \coloneqq \sum_{n=0}^{\infty} 2^{-(n+1)} \mathbb{1}_A(n)$$

is noncomputable. Indeed, a noncomputable set is neither finite nor cofinite, so this binary expansion is unique; its computability would decide A.

2 Computable Functions

Definition 2.1. Computable Function
Let $D \subseteq \mathbb{R}$. A function $f\colon D \to \mathbb{R}$ is *computable* if there exists an oracle algorithm $\mathcal{A}$ with the following property. For every $x \in D$, every function $\alpha\colon \mathbb{N}_0 \to \mathbb{Q}$ satisfying

$$|\alpha(i) - x| < 2^{-i} \qquad (i \in \mathbb{N}_0),$$

and every precision $n \in \mathbb{N}_0$, the algorithm $\mathcal{A}$, given access to α and the input n, halts and returns a rational number $\mathcal{A}(\alpha, n)$ such that

$$|\mathcal{A}(\alpha, n) - f(x)| < 2^{-n}.$$

The algorithm may query finitely many values $\alpha(i)$ during a computation. The definition extends componentwise to functions between subsets of $\mathbb{R}^k$ and $\mathbb{R}^m$.

Remark. A name α need not be computable: $\mathcal{A}$ must work for *every* rational name, even one naming a noncomputable real. Unlike a binary-digit oracle, rational approximation does not require a uniform decision at dyadic boundaries (cf. Section 1).

If $f\colon D \to \mathbb{R}$ is computable, then $f(D \cap \mathbb{R}_C) \subseteq \mathbb{R}_C$: a computable name of x can be supplied to the oracle for f.

Example 2.2. Basic Computable Functions
For $r \in \mathbb{R}_C$, both $x \mapsto x + r$ and $x \mapsto rx$ are computable. The functions $|x|$, $\exp x$, $\sin x$, $\sqrt{x}$ on $[0, \infty)$, and $\log x$ on $(0, \infty)$ are computable. In several variables, $+$, $-$, $\times$, min, and max are computable, as is division on $\{(x, y) \mid y \neq 0\}$.

Theorem 2.3. Closure under Composition
Let $f\colon D \to E$ and $g\colon E \to \mathbb{R}^m$ be computable, where $D \subseteq \mathbb{R}^k$ and $E \subseteq \mathbb{R}^\ell$. Then $g \circ f$ is computable.

Corollary 2.4. Computable Function Operations
If $f, g\colon D \to \mathbb{R}$ are computable, then so are

$$|f|, \quad f + g, \quad fg, \quad \min(f, g), \quad \max(f, g), \quad \exp \circ f.$$

Moreover, f/g is computable on $\{x \in D \mid g(x) \neq 0\}$.

Theorem 2.5. Computability Implies Continuity
Every computable function $f\colon D \to \mathbb{R}^m$, with $D \subseteq \mathbb{R}^k$, is continuous on D (with the relative topology).

Proof. Fix $\mathbf{x} \in D$ and $\varepsilon > 0$; choose $n \in \mathbb{N}_0$ with $2^{1-n} < \varepsilon$. Choose a rational name $\alpha\colon \mathbb{N}_0 \to \mathbb{Q}^k$ of $\mathbf{x}$ with slack:

$$\|\alpha(i) - \mathbf{x}\|_\infty < 2^{-(i+2)} \qquad (i \in \mathbb{N}_0).$$

At precision n, the algorithm queries a finite set of indices I. For some $\delta > 0$, every $\mathbf{y} \in D$ with $\|\mathbf{x} - \mathbf{y}\|_\infty < \delta$ also admits a name agreeing with α on I. The algorithm therefore returns the same $\mathbf{q} \in \mathbb{Q}^m$ on both names, whence

$$\|f(\mathbf{x}) - f(\mathbf{y})\|_\infty \leq \|f(\mathbf{x}) - \mathbf{q}\|_\infty + \|\mathbf{q} - f(\mathbf{y})\|_\infty < 2^{1-n} < \varepsilon.$$

Thus f is continuous at $\mathbf{x}$. □

Corollary 2.6. Discontinuous Tests
The sign function and the predicates $x < y$ and $x = y$, viewed as $\{0,1\}$-valued functions on $\mathbb{R}$ and $\mathbb{R}^2$, respectively, are not computable on their full domains.

Remark. Continuity is necessary, not sufficient, for computability: if $c \in \mathbb{R} \setminus \mathbb{R}_C$, the constant function $x \mapsto c$ is continuous but not computable. In particular, a function with a discontinuity at a point of its domain cannot be computable there.

Example 2.7. Discontinuous Functions and Their Domains
The step function $x \mapsto \mathbb{1}_{\{x>0\}}$ is not computable on $\mathbb{R}$ but is computable on $\mathbb{R} \setminus \{0\}$: refine an input approximation until its sign is certified. The same principle applies to strict comparison on $x \neq y$.

A set-valued extension, outside the single-valued definition above, can instead allow both outputs at the discontinuity:

$$\operatorname{sgn}'(x) = \begin{cases} \{1\}, & x > 0, \\ \{0\}, & x < 0, \\ \{0,1\}, & x = 0. \end{cases}$$

Theorem 2.8. Computability of the Exponential Function
The function $\exp\colon \mathbb{R} \to \mathbb{R}$ *is computable.*

Proof. First let $x \in [-\frac{1}{2}, \frac{1}{2}]$, let α be a rational name of x, and fix $n \in \mathbb{N}_0$. Set $m = d = n + 3$ and

$$\widehat{x} = \alpha(m), \qquad \widehat{y} = P_d(\widehat{x}), \qquad P_d(t) = \sum_{j=0}^{d} \frac{t^j}{j!}.$$

The rational $\widehat{y}$ is computed by finite arithmetic. Since $|x - \widehat{x}| < 2^{-m}$ and $|\widehat{x}| < 1$, the mean value theorem and $e < 3$ give

$$|e^x - e^{\widehat{x}}| < 3 \cdot 2^{-m}.$$

For $d \geq 3$, the Taylor remainder satisfies

$$|e^{\widehat{x}} - P_d(\widehat{x})| \leq \sum_{j=d+1}^{\infty} \frac{1}{j!} \leq \sum_{j=d+1}^{\infty} 2^{-j} = 2^{-d}.$$

Consequently,

$$|e^x - \widehat{y}| < (3+1)2^{-(n+3)} < 2^{-n}.$$

For arbitrary $x \in \mathbb{R}$, one can effectively choose $k \in \mathbb{N}_0$ with $|x|/2^k \leq \frac{1}{2}$. Write $u = e^{x/2^k}$ and $N = 2^k$. The preceding algorithm computes $u \in (0,2)$ to arbitrary precision. For any rational v with $|u - v| < 1$,

$$|u^N - v^N| \leq N 3^{N-1} |u - v|.$$

Thus a sufficiently accurate rational approximation v gives the required approximation v^N to $e^x = u^N$. □

Theorem 2.9. Noncomputability of Strict Comparison
The strict-comparison map $\mathbb{R}^2 \to \{0,1\}$ defined by

$$(x,y) \mapsto \begin{cases} 1, & x < y, \\ 0, & x \geq y \end{cases}$$

is not computable.

Proof. Suppose an algorithm computes strict comparison. Because its values lie in $\{0,1\}$, requesting an approximation with error below $1/4$ and testing whether the result exceeds $1/2$ yields an algorithm returning the exact bit. Give this decision algorithm the two names

$$\alpha(i) = 0, \qquad \beta(i) = 2^{-(i+1)} \qquad (i \in \mathbb{N}_0),$$

both of which represent the real number 0. Hence the algorithm must halt and output 0. During this run, only finitely many indices of β are queried; let $N \in \mathbb{N}_0$ exceed or equal all of them. Define another rational name by

$$\beta'(i) = \begin{cases} 2^{-(i+1)}, & i \le N, \\ 2^{-(N+1)}, & i > N. \end{cases}$$

For $i \le N$, its distance from $2^{-(N+1)}$ is strictly smaller than 2^{-i}, and for $i > N$ the distance is zero. Thus β' is a valid name of $y' = 2^{-(N+1)} > 0$. The algorithm cannot distinguish (α, β') from (α, β) during its finite computation; it still returns 0. But strict comparison at $(0, y')$ equals 1, a contradiction. □

Theorem 2.10. Comparison of Unequal Real Numbers
Strict comparison is computable on

$$\{(x, y) \in \mathbb{R}^2 \mid x \neq y\}.$$

Remark. If $x \neq y$, sufficiently accurate rational approximations eventually certify their order. Without this promise, neither equality nor strict comparison is decidable on all of $\mathbb{R}^2$: their indicator maps are discontinuous along $x = y$.

3 Floating-Point Arithmetic

Definition 3.1. Floating-Point Representation
Fix $E \ge 2$ and $M \ge 1$. A floating-point bit string consists of a sign bit s, E exponent bits, and M fraction bits. Let

$$k = (e_1 \cdots e_E)_2, \qquad t = (0.m_1 \cdots m_M)_2 = \sum_{j=1}^{M} m_j 2^{-j}, \qquad b_E = 2^{E-1} - 1.$$

The interpretation map $\gamma_{E,M} \colon \{0,1\}^{1+E+M} \to \mathbb{R} \cup \{-\infty, +\infty, \mathrm{NaN}\}$ is defined by

$$\gamma_{E,M}(s, e_1, \ldots, e_E, m_1, \ldots, m_M) = \begin{cases} (-1)^s 2^{k-b_E}(1+t), & 1 \le k \le 2^E - 2, \\ (-1)^s 2^{1-b_E} t, & k = 0, \\ (-1)^s \infty, & k = 2^E - 1,\ t = 0, \\ \mathrm{NaN}, & k = 2^E - 1,\ t \neq 0. \end{cases}$$

The first case represents *normal* values. The second represents *subnormal* values when $t \neq 0$ and zero when $t = 0$; the remaining cases represent infinities and *not a number* (NaN). The normal exponent bounds are

$$e_{\min} = 1 - b_E = 2 - 2^{E-1}, \qquad e_{\max} = 2^E - 2 - b_E = b_E.$$

Write

$$\overline{\mathbb{F}}_{E,M} = \operatorname{im} \gamma_{E,M}, \qquad \mathbb{F}_{E,M} = \overline{\mathbb{F}}_{E,M} \setminus \{-\infty, +\infty, \mathrm{NaN}\}.$$

Thus $\mathbb{F}_{E,M} \subseteq \mathbb{R}$ is the set of finite floating-point values.

Remark. The two signed-zero bit strings both represent $0 \in \mathbb{F}_{E,M}$, while several distinct bit strings represent NaN. The map $\gamma_{E,M}$ identifies these representations as values. In the usual binary32 and binary64 formats, $(E, M) = (8, 23)$ and $(11, 52)$, respectively.

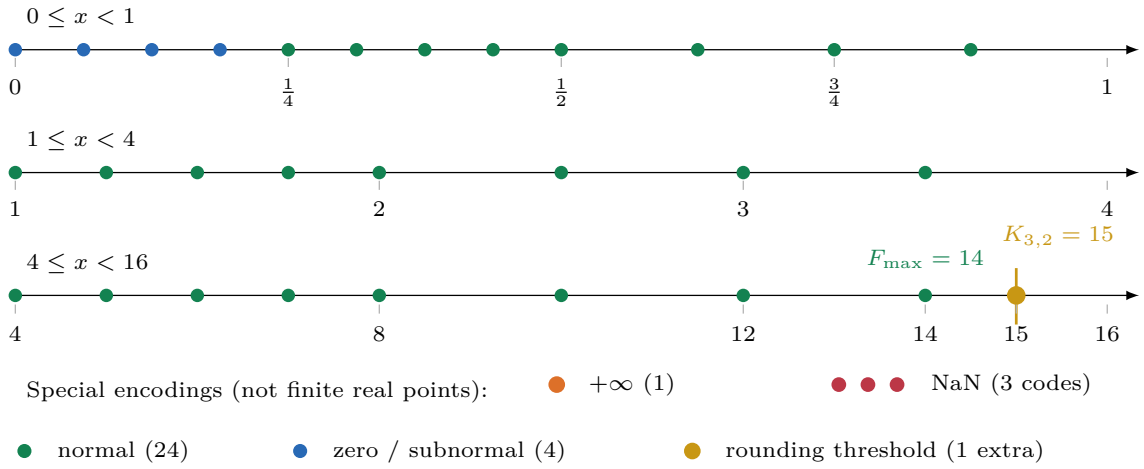


Figure 1: Nonnegative encodings for $(E, M) = (3, 2)$: 24 normal values, $+0$ and three subnormals, $+\infty$, and three NaN bit patterns. The nonrepresentable rounding threshold $K_{3,2} = 15$ is shown separately; the largest finite value is $F_{\max} = 14$.

Definition 3.2. Rounding
Let

$$F_{\max} = (2 - 2^{-M})2^{b_E}, \qquad K_{E,M} = (2 - 2^{-M-1})2^{b_E}.$$

Here $F_{\max}$ is the largest finite floating-point value and $K_{E,M}$ is the positive overflow threshold. The *round-to-nearest, ties-to-even* map $\mathrm{rnd}_{E,M} \colon \mathbb{R} \to \overline{\mathbb{F}}_{E,M}$ is defined by

$$\mathrm{rnd}_{E,M}(x) = \begin{cases} -\infty, & x \le -K_{E,M}, \\ \displaystyle\arg\min_{y \in \mathbb{F}_{E,M}} |x - y|, & |x| < K_{E,M}, \\ +\infty, & x \ge K_{E,M}. \end{cases}$$

If two finite values minimize the distance, choose the one whose least significant stored significand bit is even. In particular, halfway cases near 0 round to 0; the chosen value always exists and is unique.

Proposition 3.3. Monotonicity of Rounding
For $x, y \in \mathbb{R}$,

$$x \le y \quad \Longrightarrow \quad \mathrm{rnd}_{E,M}(x) \le \mathrm{rnd}_{E,M}(y),$$

with the usual order on $\mathbb{R} \cup \{-\infty, +\infty\}$.

Definition 3.4. Floating-Point Operations
For $a, b, c \in \mathbb{F}_{E,M}$, define rounded arithmetic by

$$a \oplus b = \mathrm{rnd}_{E,M}(a + b), \quad a \ominus b = \mathrm{rnd}_{E,M}(a - b),$$
$$a \otimes b = \mathrm{rnd}_{E,M}(ab), \qquad a \oslash b = \mathrm{rnd}_{E,M}(a/b) \quad (b \neq 0).$$

The secondary notations are $\odot$ for multiplication and $\oslash$ for division. *Fused multiply-add* performs only one rounding:

$$\mathrm{fma}(a, b, c) = \mathrm{rnd}_{E,M}(ab + c).$$

These operations have values in $\overline{\mathbb{F}}_{E,M}$; operations involving infinities, NaN, or signed-zero bit patterns are outside the present finite-input definitions.

Remark. Algebraic Properties. Rounded addition and multiplication are commutative, with identities 0 and 1, but generally neither associative nor distributive. For $(E, M) = (3, 2)$, ties-to-even gives

$$(1 \oplus \tfrac{1}{8}) \oplus (-1) = 0 \neq \tfrac{1}{8} = 1 \oplus (\tfrac{1}{8} \oplus (-1)),$$
$$2 \otimes (\tfrac{3}{4} \otimes \tfrac{1}{8}) = \tfrac{1}{4} \neq \tfrac{3}{16} = (2 \otimes \tfrac{3}{4}) \otimes \tfrac{1}{8},$$
$$3 \otimes (1 \oplus \tfrac{3}{2}) = 8 \neq 7 = (3 \otimes 1) \oplus (3 \otimes \tfrac{3}{2}).$$

Every $a \in \mathbb{F}_{E,M}$ has the unique additive inverse $-a$, since $a \oplus b = 0 \iff b = -a$ (all finite values are integer multiples of the smallest subnormal spacing). Multiplicative inverses need not exist or be unique. In the same format,

$$\tfrac{3}{2} \otimes \tfrac{5}{8} = 1 = \tfrac{3}{2} \otimes \tfrac{3}{4}, \qquad \tfrac{1}{16} \otimes b \neq 1 \quad (b \in \mathbb{F}_{3,2}).$$

Moreover, rounded operations may overflow to $\pm\infty$, outside $\mathbb{F}_{E,M}$; hence they do not form a field.

4 Error Measures and Basic Bounds

Fix the format (E, M) of Section 3. In this section write $\mathbb{F} = \mathbb{F}_{E,M}$ and $\mathrm{rnd} = \mathrm{rnd}_{E,M}$, and let $F_{\max}$, $e_{\min}$, and $e_{\max}$ retain their earlier meanings. For $x \in \mathbb{R}$, call x in the *normal range* when $2^{e_{\min}} \leq |x| \leq F_{\max}$ and in the *subnormal range* when $|x| < 2^{e_{\min}}$.

Definition 4.1. Absolute and Relative Errors
For an exact value $x \in \mathbb{R}$ and an approximation $\widehat{x} \in \mathbb{F}$, set

$$\mathrm{err}_{\mathrm{abs}}(x, \widehat{x}) \coloneqq |x - \widehat{x}|,$$
$$\mathrm{err}_{\mathrm{rel}}(x, \widehat{x}) \coloneqq \frac{|x - \widehat{x}|}{|x|} \qquad (x \neq 0).$$

Definition 4.2. Unit in the Last Place
The *unit in the last place* is the function $\mathrm{ulp}\colon \mathbb{R} \to \mathbb{R}_{>0}$ defined by

$$\mathrm{ulp}(x) \coloneqq \begin{cases} 2^{e-M}, & 2^e \leq |x| < 2^{e+1} \text{ for some integer } e \geq e_{\min}, \\ 2^{e_{\min}-M}, & |x| < 2^{e_{\min}}. \end{cases}$$

The *ULP error* of $\widehat{x} \in \mathbb{F}$ is

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) \coloneqq \frac{|x - \widehat{x}|}{\mathrm{ulp}(x)}.$$

Thus ULP error measures the discrepancy in units of the local floating-point spacing (within a common binade). The *machine epsilon* (unit roundoff) of the format is

$$\varepsilon \coloneqq \tfrac{1}{2}\,\mathrm{ulp}(1) = 2^{-M-1}.$$

Proposition 4.3. Relative and ULP Errors
Let $x \in \mathbb{R} \setminus \{0\}$ and $\widehat{x} \in \mathbb{F}$. Then

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) \leq \frac{\mathrm{err}_{\mathrm{rel}}(x, \widehat{x})}{\varepsilon}.$$

If x is in the normal range, then also

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) \geq \frac{\mathrm{err}_{\mathrm{rel}}(x, \widehat{x})}{2\varepsilon}.$$

Proof. The definitions give, without dividing by an error that might vanish,

$$\mathrm{err}_{\mathrm{ulp}}(x,\widehat{x}) = \mathrm{err}_{\mathrm{rel}}(x,\widehat{x})\frac{|x|}{\mathrm{ulp}(x)}.$$

For $2^e \le |x| < 2^{e+1}$ with $e \ge e_{\min}$,

$$2^M \le \frac{|x|}{\mathrm{ulp}(x)} < 2^{M+1} = \frac{1}{\varepsilon}.$$

In the subnormal range, the same ratio is at most 2^M; this proves the first inequality. In the normal range its lower bound is $2^M = 1/(2\varepsilon)$, proving the second.

The normal-range hypothesis cannot be dropped. Indeed, for $x = 2^{e_{\min}-M}$ and $\widehat{x} = 0$, both the relative and ULP errors equal 1, but $1/(2\varepsilon) = 2^M > 1$. □

Proposition 4.4. Rounding Error Bounds
For round-to-nearest, ties-to-even, the following estimates hold:

(i) If x is in the normal range, then

$$\mathrm{err}_{\mathrm{rel}}(x,\mathrm{rnd}(x)) \le \varepsilon.$$

(ii) If x is in the subnormal range, then

$$\mathrm{err}_{\mathrm{abs}}(x,\mathrm{rnd}(x)) \le 2^{e_{\min}}\varepsilon.$$

(iii) If $|x| \le F_{\max}$, then

$$\mathrm{err}_{\mathrm{ulp}}(x,\mathrm{rnd}(x)) \le \tfrac{1}{2}.$$

Proof. For $2^e \le |x| < 2^{e+1}$ in the normal range, consecutive floating-point values are separated by at most 2^{e-M} locally. Thus nearest rounding gives

$$\frac{|x-\mathrm{rnd}(x)|}{|x|} \le \frac{2^{e-M-1}}{2^e} = \varepsilon.$$

In the subnormal range the spacing is $2^{e_{\min}-M}$, so

$$|x-\mathrm{rnd}(x)| \le \tfrac{1}{2}2^{e_{\min}-M} = 2^{e_{\min}}\varepsilon.$$

Finally, for $|x| \le F_{\max}$, nearest rounding is within half the local spacing $\mathrm{ulp}(x)$, giving (iii). □

Definition 4.5. Underflow and Overflow
For an exact nonzero arithmetic result z, the *underflow range* is $0 < |z| < 2^{e_{\min}}$. A result beyond the finite range satisfies $|z| > F_{\max}$; under round-to-nearest, actual *overflow to infinity* occurs when $|z| \ge K_{E,M}$. These two upper thresholds need not coincide.

Theorem 4.6. Error of Floating-Point Operations
Let $a, b \in \mathbb{F}$ and choose corresponding operations

$$* \in \{+,-,\times,\div\}, \qquad \circledast \in \{\oplus,\ominus,\otimes,\oslash\},$$

with $b \ne 0$ for division. If $z = a * b$ satisfies $|z| \le F_{\max}$, then

$$a \circledast b = z(1+\delta)+\delta', \qquad |\delta| \le \varepsilon, \quad |\delta'| \le 2^{e_{\min}}\varepsilon,$$

where one may choose $\delta' = 0$ in the normal range and $\delta = 0$ in the subnormal range (including $z = 0$). For addition and subtraction, one may always choose $\delta' = 0$.

Proof. By definition, $a \circledast b = \mathrm{rnd}(z)$. In the normal range, the relative rounding bound yields $\mathrm{rnd}(z) = z(1+\delta)$ with $|\delta| \le \varepsilon$. In the subnormal range, the absolute bound yields $\mathrm{rnd}(z) = z + \delta'$ with $|\delta'| \le 2^{e_{\min}}\varepsilon$.

For $* \in \{+, -\}$, every element of $\mathbb{F}$ is an integer multiple of $2^{e_{\min}-M}$, so $a \pm b$ is also such a multiple. If this exact sum or difference is in the subnormal range, it is representable and rounding is exact. Thus $\delta' = 0$ in both ranges. □

Example 4.7. Error in Repeated Multiplication
Let $a, b, c \in \mathbb{F} \setminus \{0\}$, and suppose both intermediate exact products are in the normal range and do not overflow. By the preceding theorem, for some $|\delta_1|, |\delta_2| \le \varepsilon$,

$$\begin{aligned}(a \otimes b) \otimes c &= \big(ab(1+\delta_1)c\big)(1+\delta_2) \\ &= abc(1+\delta_1)(1+\delta_2).\end{aligned}$$

Consequently, with $x = abc$ and $\widehat{x} = (a \otimes b) \otimes c$,

$$\begin{aligned}\mathrm{err}_{\mathrm{rel}}(x, \widehat{x}) &\le |\delta_1| + |\delta_2| + |\delta_1\delta_2| \le 2\varepsilon + \varepsilon^2, \\ \mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) &\le 2 + \varepsilon.\end{aligned}$$

To first order the relative error is at most 2ε.

Example 4.8. Error in Repeated Addition
Let $a, b, c \in \mathbb{F}$ and set $s = a + b$, $u = a + b + c \neq 0$. Assume $|s|, |(a \oplus b) + c| \le F_{\max}$. For rounded addition, the theorem gives $\delta' = 0$ even for subnormal results; hence

$$\begin{aligned}(a \oplus b) \oplus c &= \big((a+b)(1+\delta_1) + c\big)(1+\delta_2) \\ &= u(1+\delta_2) + s\delta_1(1+\delta_2), \qquad |\delta_1|, |\delta_2| \le \varepsilon.\end{aligned}$$

Writing $x = u$ and $\widehat{x} = (a \oplus b) \oplus c$, we obtain

$$\begin{aligned}\mathrm{err}_{\mathrm{rel}}(x, \widehat{x}) &\le \varepsilon \left(1 + \frac{|a+b|}{|a+b+c|}\right) + \varepsilon^2 \frac{|a+b|}{|a+b+c|}, \\ \mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) &\le 1 + \frac{|a+b|}{|a+b+c|} + \varepsilon \frac{|a+b|}{|a+b+c|}.\end{aligned}$$

The ratio $|a+b|/|a+b+c|$ can amplify the first rounding error.

For example, when $2^{M+1} \in \mathbb{F}$, take $a = 1$, $b = 2^{M+1}$, and $c = -2^{M+1}$. Ties-to-even yields

$$(a \oplus b) \oplus c = 0, \qquad a + b + c = 1,$$

with relative error 1 and ULP error 2^M.

5 Catastrophic Cancellation and Stable Reformulations

Remark. Catastrophic Cancellation. When large, nearly equal quantities cancel, an error previously negligible relative to the intermediate values can dominate the final result. The preceding example loses the small term a in the first rounded addition.

The same phenomenon can reveal *pre-existing input errors*. If $\widehat{x}, \widehat{y} \in \mathbb{F}$ approximate $x, y \in \mathbb{R}$, then even when their subtraction is exact,

$$|(\widehat{x} - \widehat{y}) - (x - y)| \le |\widehat{x} - x| + |\widehat{y} - y|.$$

For $x \neq y$, dividing by $|x - y|$ gives the relative error; dividing by $\mathrm{ulp}(x - y)$ gives the ULP error. Both can be large when $x \approx y$, even if the subtraction itself is exact.

Example 5.1. Stable Evaluation of $a^2 - b^2$
For $a, b \in \mathbb{F}$, $a^2 \neq b^2$, compare

$$u = (a \otimes a) \ominus (b \otimes b), \qquad v = (a \ominus b) \otimes (a \oplus b).$$

Assume that the operations do not overflow or underflow and that the usual relative-error model applies to all intermediate results. Writing $|\delta_i| \leq \varepsilon$ and retaining terms through first order gives

$$\begin{aligned} u &= (a^2(1+\delta_1) - b^2(1+\delta_2))(1+\delta_3), \\ \frac{|u - (a^2 - b^2)|}{|a^2 - b^2|} &\leq \left(1 + \frac{a^2 + b^2}{|a^2 - b^2|}\right)\varepsilon + O(\varepsilon^2), \\ v &= (a-b)(a+b)(1+\delta_1)(1+\delta_2)(1+\delta_3), \\ \frac{|v - (a^2 - b^2)|}{|a^2 - b^2|} &\leq 3\varepsilon + O(\varepsilon^2). \end{aligned}$$

For a normal exact result, the corresponding ULP-error bounds are $1 + (a^2 + b^2)/|a^2 - b^2| + O(\varepsilon)$ and $3 + O(\varepsilon)$, respectively. The first can be arbitrarily large when $|a| \approx |b|$; the factored expression avoids subtracting two rounded squares.

Remark. For $r, r' \in \mathbb{R}$ with $\mathrm{err}_{\mathrm{ulp}}(r, r') \leq 1$ and finite $\mathrm{rnd}(r')$,

$$\mathrm{err}_{\mathrm{ulp}}(r, \mathrm{rnd}(r')) \leq \mathrm{err}_{\mathrm{ulp}}(r, r') + \tfrac{1}{2}.$$

Indeed, if $\mathrm{ulp}(r') \leq \mathrm{ulp}(r)$, the triangle inequality and nearest rounding suffice. Otherwise r' lies across a power-of-two boundary from r. The one-ULP hypothesis forces r' to round back to that boundary, closer to r than r' was. The estimate thus holds in both cases.

Example 5.2. Stable Quadratic Roots
For $a \neq 0$ and $\Delta = b^2 - 4ac \geq 0$, the usual quadratic formula is

$$x_{\pm} = \frac{-b \pm \sqrt{\Delta}}{2a}.$$

If $b < 0$ and $b^2 \gg |4ac|$, evaluating $(-b - \sqrt{\Delta})/(2a)$ may suffer cancellation. Rationalization gives

$$\frac{-b - \sqrt{\Delta}}{2a} = \frac{2c}{-b + \sqrt{\Delta}},$$

provided the denominator is nonzero. The analogous rationalization applies to the other root when $b > 0$. Moreover, forming b^2 can overflow even if the roots are finite. For $b \neq 0$, the identity

$$\sqrt{b^2 - 4ac} = |b|\sqrt{1 - \frac{4ac}{b^2}}$$

illustrates a rescaling that avoids directly squaring a large b; its implementation must also guard the intermediate quotient. When $|4ac/b^2| \ll 1$, the Taylor approximation $\sqrt{1-t} = 1 - \frac{1}{2}t + O(t^2)$ explains the sensitivity of the small root. In particular, for $b^2 \gg |4ac|$,

$$x_- \approx \begin{cases} -b/a + c/b, & b > 0, \\ -c/b, & b < 0. \end{cases}$$

Higher-order terms follow from $\sqrt{1-t} = 1 - \frac{1}{2}t - \frac{1}{8}t^2 + O(t^3)$.

Example 5.3. Two Ways to Compute a Mean
For $a, b \in \mathbb{F}$ compare

$$\widehat{m} = (a \oplus b) \oslash 2, \qquad \widehat{m}' = (a \oslash 2) \oplus (b \oslash 2).$$

If $|a + b| \le F_{\max}$, the first is correctly rounded:

$$\widehat{m} = \mathrm{rnd}\left(\frac{a+b}{2}\right), \qquad \min(a, b) \le \widehat{m} \le \max(a, b).$$

It may nevertheless overflow when $|a + b|$ exceeds the finite range, even though the mean is representable. The second avoids such intermediate overflow; it is correctly rounded when the inputs are sufficiently large, and otherwise has ULP error at most 1.

Theorem 5.4. Correctly Rounded Mean
Let $C \in [2^{e_{\min}+M+3}, 2^{e_{\max}}] \cap \mathbb{F}$. For $a, b \in \mathbb{F}$, define

$$m(a, b) = \begin{cases} (a \oplus b) \oslash 2, & |a| < C \text{ and } |b| < C, \\ (a \oslash 2) \oplus (b \oslash 2), & \text{otherwise.} \end{cases}$$

Then, assuming the stated floating-point operations are defined,

$$m(a, b) = \mathrm{rnd}\left(\frac{a+b}{2}\right) \qquad (a, b \in \mathbb{F}).$$

6 Error-Free Computations

Fix $\mathbb{F} = \mathbb{F}_{E,M}$ and $\mathrm{rnd} = \mathrm{rnd}_{E,M}$ as before. For $a \in \mathbb{F} \setminus \{0\}$ let $e_a \in [e_{\min}, e_{\max}] \cap \mathbb{Z}$ denote its floating-point exponent, taking $e_a = e_{\min}$ for subnormals. Set $e_0 = e_{\min}$ when an exponent is needed for zero. Write $\Delta_{\min} \coloneqq 2^{e_{\min}-M}$ for the least positive subnormal spacing. All exactness assertions below concern finite results with the indicated range conditions.

Lemma 6.1. Integer Significands
A real number a is in $\mathbb{F}$ if and only if there exist $m, e \in \mathbb{Z}$ with

$$a = m2^{e-M}, \qquad |m| < 2^{M+1}, \qquad e_{\min} \le e \le e_{\max}.$$

In particular, every $a \in \mathbb{F}$ is an integer multiple of $\Delta_{\min}$.

Proof. For normal values take $e = e_a$ and let m be the integer encoded by the $(M+1)$ significand bits; for subnormals take $e = e_{\min}$ and $|m| < 2^M$. Conversely, for any displayed pair (m, e), a significand with at most $M + 1$ bits is either normal at a suitable exponent or subnormal after shifting to $e_{\min}$. If a shift would exceed $e_{\max}$, the bound $|m| < 2^{M+1}$ prevents it. □

Theorem 6.2. Hauser: Exact Addition
For $a, b \in \mathbb{F}$,

$$|a + b| \le 2^{e_{\min}+1} \implies a \oplus b = a + b.$$

Proof. Both a and b are integer multiples of $\Delta_{\min}$, hence so is $a + b$. Every such multiple in $[-2^{e_{\min}+1}, 2^{e_{\min}+1}]$ is representable. Therefore $a + b \in \mathbb{F}$ and rounding does nothing. □

Theorem 6.3. Sterbenz: Exact Subtraction
For $a, b \in \mathbb{F}$,

$$\frac{a}{2} \le b \le 2a \implies a \ominus b = a - b.$$

Equivalently, one may assume $b/2 \le a \le 2b$.

Proof. The hypothesis forces $a, b \geq 0$. The zero case is immediate. By exchanging a, b and changing the sign if necessary, assume $a \geq b$, so $b \leq a \leq 2b$. Write $a = m_a 2^{e_a - M}$ and $b = m_b 2^{e_b - M}$ by the lemma. As $a \geq b$, we have $e_a \geq e_b$ (with the convention for subnormals), and thus

$$a - b = (m_a 2^{e_a - e_b} - m_b) 2^{e_b - M} = m' 2^{e_b - M}, \qquad m' \in \mathbb{Z}.$$

Since $0 \leq a - b \leq b$, one has $0 \leq m' \leq m_b < 2^{M+1}$, so $a - b \in \mathbb{F}$. Thus $a \ominus b = a - b$. □

Proposition 6.4. Exact Scaling by Powers of Two
Suppose $a, b \in \mathbb{F} \setminus \{0\}$ have $|a| = (\alpha_0.\alpha_1 \ldots \alpha_M)_2 \, 2^{e_a}$ and $|b| = 2^{e_b}$. If $e_a + e_b \in [e_{\min}, e_{\max}]$, then

$$a \otimes b = ab.$$

Similarly, if $e_a - e_b \in [e_{\min}, e_{\max}]$, then $a \oslash b = a/b$.

Proposition 6.5. Exact Products with Trailing Zeros
Let $a, b \in \mathbb{F}$ be nonzero normal values whose binary significands have, respectively, at least j and j' trailing zero bits. If

$$j + j' \geq M + 1, \qquad e_{\min} \leq e_a + e_b \leq e_{\max} - 1,$$

then $ab \in \mathbb{F}$ and $a \otimes b = ab$.

Proof. Removing the trailing zeros leaves integer significands of at most $M + 1 - j$ and $M + 1 - j'$ bits. Their product has at most $2(M + 1) - j - j' \leq M + 1$ bits. The exact product therefore fits the available significand precision; its normalized exponent is either $e_a + e_b$ or $e_a + e_b + 1$, both representable by hypothesis. □

Remark. Sterbenz's theorem does not preclude catastrophic cancellation: subtraction of nearby *represented* floats may be exact, while an error inherited from earlier rounded operations becomes large relative to the small difference.

Theorem 6.6. Dekker's Theorem (Fast2Sum)
For $a, b \in \mathbb{F}$ with $|a + b| \leq F_{\max}$, put

$$s = a \oplus b, \qquad r = (a + b) - s.$$

Then $r \in \mathbb{F}$. If $e_a \geq e_b$, the algorithm

$$\begin{aligned} s &= a \oplus b, \\ b' &= s \ominus a, \qquad \textit{(Fast2Sum)} \\ z &= b \ominus b' \end{aligned}$$

returns $(s, z) = (a \oplus b, r)$; in particular, $a + b = s + z$ exactly.

Proof. By symmetry assume $e_a \geq e_b$. We first show $s - a \in \mathbb{F}$. If a, b have the same sign and $|a| \geq |b|$, then s has that sign and $|a| \leq |s| \leq 2|a|$; Sterbenz's theorem applies to $s - a$. If their signs differ and $|b| \geq |a|/2$, their magnitudes differ by at most a factor of two, so Sterbenz makes $a + b$ exact. If their signs differ and $|b| < |a|/2$, either the sum is exact in the subnormal range by Hauser's theorem, or s and a are within a factor of two and Sterbenz applies again. In the remaining case, a, b have the same sign with $|b| > |a|$, which forces $e_a = e_b$. Both belong to the same binade of spacing $h = 2^{e_a - M}$; $s - a$ is a multiple of h with magnitude at most $2^{e_a + 1}$, hence is representable. Thus $b' = s \ominus a = s - a$.

Moreover, a, b, and s lie on the grid $2^{e_b - M}\mathbb{Z}$; if the exact sum is smaller than 2^{e_b}, it is itself representable, and otherwise rounding uses a no-finer grid. Consequently, $r = b - b'$ is an integer multiple of $2^{e_b - M}$. By optimality of nearest rounding, comparison with the representable candidate a yields

$$|r| = |(a + b) - s| \leq |(a + b) - a| = |b|.$$

The integer-significand lemma now implies $r \in \mathbb{F}$. Therefore $b \ominus b' = r$, giving $a + b = s + r$ exactly. □

Theorem 6.7. Boldo's Theorem (Product Residual)
For $a, b \in \mathbb{F}$ with $|ab| \le F_{\max}$, let $s = a \otimes b$ and $r = ab - s$. If

$$e_a + e_b \ge e_{\min} + M,$$

then $r \in \mathbb{F}$ and

$$r = \mathrm{fma}(a, b, -s).$$

Proof. In the integer-significand representation, $ab = m_a m_b 2^{e_a+e_b-2M}$. The rounded product s lies on the same binary grid, so $ab - s = k2^{e_a+e_b-2M}$ for some integer k. The exact product has exponent $e_a + e_b$ or $e_a + e_b + 1$; nearest rounding therefore gives $|k| \le 2^M$. By the hypothesis, $e_a + e_b - M \ge e_{\min}$, so the residual has a representable significand and exponent. Thus $r \in \mathbb{F}$, and the single-rounding property of FMA gives $\mathrm{fma}(a, b, -s) = \mathrm{rnd}(r) = r$. □

7 Automatic Error Analysis

Definition 7.1. Floating-Point Expression Syntax
Fix $N \in \mathbb{N}$ and $\mathbb{F} = \mathbb{F}_{E,M}$. Expressions are formed according to

$$e ::= c \mid x_i \mid -e \mid e + e \mid e \times e, \qquad c \in \mathbb{F}, \quad 1 \le i \le N.$$

An expression e is a syntactic object, not a function.

Definition 7.2. Floating-Point Semantics
For an input valuation $\sigma = (\sigma_1, \ldots, \sigma_N) \in \mathbb{F}^N$, the floating-point semantics is the function $[\![e]\!] \colon \mathbb{F}^N \to \mathbb{F}$ defined by

$$\begin{aligned} [\![c]\!](\sigma) &= c, & [\![x_i]\!](\sigma) &= \sigma_i, \\ [\![-e]\!](\sigma) &= -[\![e]\!](\sigma), & [\![e_1 + e_2]\!](\sigma) &= [\![e_1]\!](\sigma) \oplus [\![e_2]\!](\sigma), \\ [\![e_1 \times e_2]\!](\sigma) &= [\![e_1]\!](\sigma) \otimes [\![e_2]\!](\sigma). \end{aligned}$$

The interpretation is defined for input valuations at which every rounded operation has a finite result.

Definition 7.3. Real-Valued Semantics
The real-valued interpretation $[\![e]\!]_{\mathrm{r}} \colon \mathbb{F}^N \to \mathbb{R}$ is obtained from the same recursive rules by replacing $\oplus$ and $\otimes$ with exact $+$ and $\times$. Thus $[\![e]\!]_{\mathrm{r}}(\sigma)$ evaluates e over $\mathbb{R}$ at the same input σ.

Example 7.4. Different Semantics
Let $e = (1 + x_1) + (-x_1)$, and assume $2^{M+1} \in \mathbb{F}$. At $\sigma_1 = 2^{M+1}$, ties-to-even yields

$$[\![e]\!](\sigma) = 0, \qquad [\![e]\!]_{\mathrm{r}}(\sigma) = 1.$$

Definition 7.5. Sound Range and Error Analysis
Fix the admissible inputs

$$\Sigma := \prod_{i=1}^{N} ([l_i, r_i] \cap \mathbb{F}), \qquad l_i, r_i \in \mathbb{F}, \quad l_i \le r_i.$$

A sound *range analysis* assigns an interval $\mathrm{range}(e)$, and a sound *absolute-error analysis* assigns $\mathrm{err}(e) \ge 0$, such that

$$\begin{aligned} [\![e]\!](\sigma) &\in \mathrm{range}(e), \\ |[\![e]\!](\sigma) - [\![e]\!]_{\mathrm{r}}(\sigma)| &\le \mathrm{err}(e) \end{aligned} \qquad (\forall \sigma \in \Sigma).$$

These bounds are computed recursively from the syntax and the rounding estimates; the fixed input set Σ is suppressed in the notation.

Definition 7.6. Recursive Range Bounds
Set

$$\operatorname{range}(c) = [c, c], \qquad \operatorname{range}(x_i) = [l_i, r_i].$$

If $\operatorname{range}(e) = [a, b]$, put $\operatorname{range}(-e) = [-b, -a]$. If $\operatorname{range}(e_i) = [a_i, b_i]$ for $i = 1, 2$, put

$$\operatorname{range}(e_1 + e_2) = [\operatorname{rnd}(a_1 + a_2), \operatorname{rnd}(b_1 + b_2)].$$

Proposition 7.7. Soundness of Recursive Range Bounds
The preceding rules enclose each floating-point evaluation, provided the rounded endpoint operations remain finite.

Proof. Constants, variables, and negation follow directly from the semantics. If $[\![e_i]\!](\sigma) \in [a_i, b_i]$, then

$$a_1 + a_2 \le [\![e_1]\!](\sigma) + [\![e_2]\!](\sigma) \le b_1 + b_2.$$

Monotonicity of rnd gives the asserted range enclosure for the sum. Structural induction completes the proof. □

Definition 7.8. Recursive Absolute-Error Bounds
Set

$$\operatorname{err}(c) = \operatorname{err}(x_i) = 0, \qquad \operatorname{err}(-e) = \operatorname{err}(e).$$

If $\operatorname{range}(e_i) = [a_i, b_i]$ for $i = 1, 2$, define

$$\operatorname{err}(e_1 + e_2) = \operatorname{err}(e_1) + \operatorname{err}(e_2) + \tfrac{1}{2}\operatorname{ulp}(\max\{|a_1 + a_2|, |b_1 + b_2|\}).$$

Proposition 7.9. Soundness of Recursive Error Bounds
The preceding recursion bounds the absolute difference of the two semantics, provided the rounded operations remain finite.

Proof. Write $u_i = [\![e_i]\!](\sigma)$ and $v_i = [\![e_i]\!]_{\mathrm{r}}(\sigma)$. Then

$$\begin{aligned} |\operatorname{rnd}(u_1 + u_2) - (v_1 + v_2)| &\le |\operatorname{rnd}(u_1 + u_2) - (u_1 + u_2)| + |u_1 - v_1| + |u_2 - v_2| \\ &\le \tfrac{1}{2}\operatorname{ulp}(u_1 + u_2) + \operatorname{err}(e_1) + \operatorname{err}(e_2). \end{aligned}$$

Since $u_i \in [a_i, b_i]$ and ulp is nondecreasing in $|x|$,

$$\operatorname{ulp}(u_1 + u_2) \le \operatorname{ulp}(\max\{|a_1 + a_2|, |b_1 + b_2|\}).$$

The other recursive cases are immediate. □

Definition 7.10. Multiplication Bounds
Suppose $\operatorname{range}(e_i) = [a_i, b_i]$ for $i = 1, 2$. Let

$$p_- \coloneqq \min\{a_1a_2, a_1b_2, b_1a_2, b_1b_2\}, \qquad p_+ \coloneqq \max\{a_1a_2, a_1b_2, b_1a_2, b_1b_2\}.$$

Define

$$\operatorname{range}(e_1 \times e_2) = [\operatorname{rnd}(p_-), \operatorname{rnd}(p_+)].$$

For $i = 1, 2$, write $U_i = \max\{|a_i|, |b_i|\}$ and $\eta_i = \operatorname{err}(e_i)$. Set

$$\begin{aligned} \operatorname{err}(e_1 \times e_2) = U_1\eta_2 + U_2\eta_1 + \eta_1\eta_2 \\ + \tfrac{1}{2}\operatorname{ulp}(\max\{|p_-|, |p_+|\}). \end{aligned}$$

Proposition 7.11. Soundness of Multiplication Bounds
The multiplication rules are sound if the corresponding rounding operations have finite results.

Proof. Let $u_i = [\![e_i]\!](\sigma)$ and $v_i = [\![e_i]\!]_{\mathrm{r}}(\sigma)$. Since $u_i \in [a_i, b_i]$, the exact product u_1u_2 belongs to $[p_-, p_+]$. Rounding monotonicity proves the range enclosure. For the error, $|u_i| \le U_i$ and $|v_i| \le U_i + \eta_i$ give

$$\begin{aligned} |\operatorname{rnd}(u_1u_2) - v_1v_2| &\le |\operatorname{rnd}(u_1u_2) - u_1u_2| + |u_1u_2 - v_1v_2| \\ &\le \tfrac{1}{2}\operatorname{ulp}(u_1u_2) + U_1\eta_2 + (U_2 + \eta_2)\eta_1. \end{aligned}$$

The first term is bounded by $\frac{1}{2}\operatorname{ulp}(\max\{|p_-|, |p_+|\})$. □

Definition 7.12. Square-Root Extension
The expression grammar may be extended by $\sqrt{e}$, on input valuations for which both interpretations of e are nonnegative:

$$\begin{aligned} [\![\sqrt{e}]\!](\sigma) &= \operatorname{rnd}\big(\sqrt{[\![e]\!](\sigma)}\big), \\ [\![\sqrt{e}]\!]_{\mathrm{r}}(\sigma) &= \sqrt{[\![e]\!]_{\mathrm{r}}(\sigma)}. \end{aligned}$$

If $\operatorname{range}(e) = [a, b] \subseteq [0, F_{\max}]$, set

$$\begin{aligned} \operatorname{range}(\sqrt{e}) &= [\operatorname{rnd}(\sqrt{a}), \operatorname{rnd}(\sqrt{b})], \\ \operatorname{err}(\sqrt{e}) &= \sqrt{\operatorname{err}(e)} + \tfrac{1}{2}\operatorname{ulp}(\sqrt{b}). \end{aligned}$$

Proposition 7.13. Soundness of the Square-Root Bounds
The preceding extension is sound whenever its real and floating-point square roots are defined and finite.

Proof. Monotonicity gives the range enclosure. For $u, v \ge 0$, $|\sqrt{u} - \sqrt{v}| \le \sqrt{|u - v|}$; hence

$$\begin{aligned} |\operatorname{rnd}(\sqrt{u}) - \sqrt{v}| &\le |\operatorname{rnd}(\sqrt{u}) - \sqrt{u}| + |\sqrt{u} - \sqrt{v}| \\ &\le \tfrac{1}{2}\operatorname{ulp}(\sqrt{b}) + \sqrt{\operatorname{err}(e)}. \end{aligned}$$

□

Remark. Limitations of Interval Analysis. Independent bounds lose dependencies between repeated occurrences of the same variable. For $e = x_1 - x_1$ (i.e., $x_1 + (-x_1)$) and $x_1 \in [-1, 1]$, interval propagation gives $[-2, 2]$ although both semantics evaluate to 0. Even a sound error bound can be positive when the actual rounding error vanishes; soundness does not imply sharpness.